PROJECT	Credit Card Approval Prediction	TEAM	AI/ML Program
DOCUMENT	**Project Documentation**	DATE	July 2026

Project Documentation

Project Summary

The Credit Card Approval Prediction system automates credit card approval decisions using machine learning. Applicant financial and demographic data, combined with historical credit repayment behavior, is used to train and compare four classification algorithms. The best-performing model, Random Forest, is integrated into a Flask web application that provides real-time approval predictions.

Dataset

- application_record.csv - 438,557 applicant records, 18 columns
- credit_record.csv - 1,048,575 monthly credit status records, 3 columns
- Merged and cleaned dataset used for training - 36,457 applicants, 17 features + target

Pipeline Summary

- Epic 1: Data Collection - dataset sourced from Kaggle via the Kaggle API
- Epic 2: Visualization & Analysis - univariate, multivariate, and descriptive analysis
- Epic 3: Data Preprocessing - cleaning, merging, feature engineering, encoding
- Epic 4: Model Building - four algorithms trained, Random Forest selected
- Epic 5: Application Building - Flask web app with home, form, and result pages

Repository

Full source code, trained model, and documentation are available at: https://github.com/thePragna/credit-card-approval-prediction